

Common Formulae

Poisson / Normal approximation

$np < 10, p \text{ small} \implies$ Poisson approximation

$np, n(1-p) > 10 \implies$ Normal approximation

CLT

$X_{0 < i < n}$ independent RVs.

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Markov's Inequality

For any nonnegative random variable X and $a > 0$

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Chebychev's Inequality

Let X be a random variable and let $g(x)$ be a non-negative function. Then, for any $r > 0$

$$P(g(X) \geq r) \leq \frac{Eg(X)}{r}$$

Normal Probability Inequality

With Z as a normal distribution,

$$P(|Z| \geq t) \leq \sqrt{\frac{e}{\pi}} \frac{e^{-t^2/2}}{t} \text{ for all } t > 0$$

Kolmogorov's Inequality

Assuming independent RV S_k with finite variance

$$P\left(\max_{1 \leq k \leq n} |S_k| \geq \lambda\right) \leq \frac{\text{Var}(S_n)}{\lambda^2}$$

Expectation

$$E[g(X)] = \int g(x) f_X(x) dx$$

Expectation is linear.

Joint Expectation

$$E[g(X, Y)] = \int \int g(x, y) f_{X,Y}(x, y) dx dy$$

Moments

$$\mu'_n = \mathbb{E}[X^n]$$

$$\mu_n = \mathbb{E}[(X - \mathbb{E}[X])^n]$$

$$\mu = \mu'_1$$

$$\text{Var} = \mu_2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$M_X(t) = \mathbb{E}[e^{tX}]$$

$$\mu'_n = M_X^{(n)}(0)$$

$$M_{X,Y}(s, t) = \mathbb{E}[e^{sX+tY}]$$

$$\mathbb{E}[X^m Y^n] = \left. \frac{\partial^{m+n}}{\partial s^m \partial t^n} M_{X,Y}(s, t) \right|_{0,0}$$

$$\text{Independent} \iff M_{X,Y}(s, t) = M_X(s)M_Y(t)$$

Covariance & Correlation

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

$$\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y)$$

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$\text{Independent} \implies \rho = 0$$

Variance

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$